Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/25/2005
PI Name:	Gewirtz, Alan M. M.D.		
Project Title:	Effect of Deep Space Radiation on Human Hematopoietic Stem and Progenitor Cell Function		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Immunology, Infection and Hematology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104	**Congressional District:**	2
Comments:	PI passed away in 2010.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	07/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sutherland, Betsy (Brookhaven National Laboratory)		
Grant/Contract No.:	NCC 9-58-IIH00404		
Performance Goal No.:			
Performance Goal Text:			
	Astronauts on long term missions in deep space will be placed at risk from a variety of hazards. Some of these are known while others may be anticipated. Damage to hematopoietic stem cells as a result of radiation exposure is as an example of the latter. Our long term goal is to identify and quantitate the risks of deep space radiation to the human hematopoietic system, with particular emphasis on the hematopoietic stem cell. Stem cells are the ultimate source of both the blood and immune systems and damage to these cells could have grave immediate, and long term consequences. At the same time, because these cells can be readily removed from the body, manipulated, and stored, they are also unique candidates for countermeasures that might obviate, or totally negate, damage incurred to them. Accordingly, this project will have three specific aims which support our long term goal and these are to: 1] Investigate the cellular consequences of exposing human hematopoietic stem (HSC)and progenitor (HPC)cells to an environment which simulates the radiation environment of deep space; 2] Examine the molecular consequences of exposing human		

Task Description:	hematopoietic stem cells to an environment which simulates the radiation environment of deep space. This aim has two purposes. If radiation leads to degradation of hematopoietic cell function it will clearly be of interest to look for the molecular lesions potentially responsible for such damage. Alternatively, more long term, but initially occult damage may also be induced. The consequences of such damage could lead either to a complete failure of hematopoiesis (aplastic anemia) or the development of hematologic malignancies. Identification of such damage is therefore important; and 3] Design potential countermeasures to obviate or negate cellular and molecular damage discerned during the course of carrying out Aims 1 and 2. We propose both simple and more complex solutions to problems that might be identified during the course of this study. We suggest that prophylactic (pre-flight) harvest and storage of astronaut stem cells might be a safe, effective, and relatively inexpensive mechanism for countering long term damage to cells of the hematopoietic systems. Countermeasures which might prove effective in combating damage encountered during flight will also be developed and explored for their utility.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of effective radiation countermeasures could have significant impact on earth for civilians and military personnel alike. With regard to the civilian population, it is quite conceivable that results we obtain will be relevant to patients undergoing cancer chemotherapy and radiation therapy. In this regard it is possible that our studies will provide reagents and strategies for helping to protect normal tissues from the collateral damage of anticancer treatments. It is also possible that results we generate will be relevant to radiation workers, and members of the armed forces who may be exposed to radiation during the course of carrying our their respective duties.
Task Progress:	Progress is detailed in the Main Findings section.
Bibliography Type:	Description: (Last Updated: 04/12/2011)
Articles in Peer-reviewed Journals	PA UL A V. BE NN ET T, * NE LA S. CI NT RO N, * LA UR EN T GR OS, y JA CQ UEs LA VA L, y and BE TS Y M. SU TH ER LA ND* "ARE ENDOGENOUS CLUSTERED DNA DAMAGES INDUCED IN HUMAN CELLS?" N/A , Jan-2004
Awards	N/A "Appointed first C. Willard Robinson Professor of Hematology/Oncology, University of Pennsylvania School of Medicine" Jan-2004
Presentation	Gurudutta U Gangenahalli, Ph.D. 1, John M Millholland, Ph.D. 1, Anna Kalota, M. S. "Countering Low-Linear Energy Transfer (LET) and High-LET Radiation Exposures With Pro-Methyl Hoechst (PMH) and EUK-134 (Mg-Salen Complex): Radioprotection of Normal Human CD34+ Cells in a Simulated Deep Space Environment" N/A May-2005
Presentation	Gurudutta U Gangenahalli, Ph.D. 1, John M Millholland, Ph.D. 1, Anna Kalota, M. S. "Countering Low-Linear Energy Transfer (LET) and High-LET Radiation Exposures With Pro-Methyl Hoechst (PMH) and EUK-134 (Mg-Salen Complex): Radioprotection of Normal Human CD34+ Cells in a Simulated Deep Space Environment" N/A Dec-2004